CSE 318: Artificial Intelligence Sessional

Online 01

A2

N-Puzzle with Locked Tiles

Problem Statement

Modify your existing N-Puzzle solver so that some tiles are **locked**. A locked tile cannot be moved from its current position.

The goal state remains the standard goal configuration:

```
1 2 3 ... n 0
```

where $n = k^2 - 1$.

However, during the search, the blank tile cannot swap with any locked tile.

Additional Rule: Auto-Locking

In addition to the initially locked tiles, a tile becomes locked automatically if it is currently in its correct goal position.

That means, during neighbor generation, the blank tile cannot swap with a tile if that tile is already in its goal position.

For example, in a 3×3 puzzle, the standard goal position of tile `5` is the center cell. If tile `5` is currently in the center cell, then tile `5` cannot be moved anymore.

Input Format

The input format is:

```
k
m
locked tile values
initial board
```

Here, k is the board size and m is the number of locked tiles.

Sample Input

```
3
1
5
4 1 2
7 5 3
0 8 6
```

Sample Output

```
Minimum number of moves = 6

Step 0:
4 1 2
7 5 3
0 8 6

Step 1:
4 1 2
0 5 3
7 8 6

Step 2:
0 1 2
4 5 3
7 8 6

Step 3:
1 0 2
4 5 3
7 8 6

Step 4:
1 2 0
4 5 3
7 8 6

Step 5:
1 2 3
4 5 0
7 8 6
```

```
Step 6:
1 2 3
4 5 6
7 8 0
```

Assumption

For this task:

- All locked tiles will already be in their correct goal positions.
- All given test cases will be solvable under the locked-tile restriction.